

Khush A. Patel

Harrisburg, Pennsylvania

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Objective

First-year Mechanical Engineering student interested in robotics, control systems, and aerodynamic–mechanical design, seeking opportunities to support robotic control, or multiphysics simulation.

Education

Johns Hopkins University, Whiting School of Engineering

Baltimore, MD

B.S. in Mechanical Engineering

Anticipated May 2029

Relevant Coursework: Intro to CAD and Design (SolidWorks), Mechanics, Python, Digital Electronics, Electricity and Magnetism, Multivariable Calculus, Oral Presentations, Intro to CAD (Creo), Neural Networks From Scratch

Technical Skills

- **Software:** SolidWorks, ANSYS Workbench, Fluent, CFX, Mechanical, ROS 2, Webots, Helioscope, Microsoft 365
- **Programming:** Python (NumPy, SciPy, Matplotlib), MATLAB, R, FFT analysis
- **Simulation:** Fluid–Structure Interaction (one-way and two-way), transient CFD, turbulence modeling, static FEA
- **Prototyping:** 3D printing, laser cutting, soldering, microcontrollers, mechanical assembly, basic electronics

Engineering Experience

Amarkosha Wind Harvester Start-Up

Baltimore, MD

Mechanical Lead, Product Development

Jul 2025–Present

- Leading **VIV harvester** design and multiphysics simulation in SolidWorks and ANSYS.
- Performed two-way transient **FSI analysis** (Fluent/CFX + Structural) to assess aerodynamic–structural coupling.
- Applied **FFT-based signal processing** in MATLAB to extract vortex shedding and lock-in behavior.
- Produced detailed **dimensioned mechanical drawings** for manufacturing and prototyping.
- Accepted to **SPARK Accelerator Program** and selected as **BII Grant finalist**.

Johns Hopkins Mars Rover Team – Mechanical Subteam

Baltimore, MD

Mechanical & Controls Member (Wrist Mechanism)

Sep 2025–Present

- Designed a **3-DOF wrist** to enhance manipulator dexterity and payload handling.
- Conducted **torque analysis** for motor and joint optimization.
- Helped design a **cycloidal gearbox**, focusing on the **housing, motor mount, and encoder housing**.
- Using ROS2 for robotic manipulator testing.

GreenWorks Development

Mechanicsburg, PA

Technical Sales & Project Intern

Jan–Aug 2025

- Modeled solar systems in Helioscope and Excel for a \$5M performance validation project pitch, contributing to proposal presentations with corporate leadership.

UPMC Magee–Women’s Hospital

Pittsburgh, PA

Research Assistant, Dept. of Anesthesiology

Jun–Sep 2024

- Processed and visualized clinical datasets in R to support pre-eclampsia diagnostics research.

Projects

4-DOF Robotic Arm (Independent)

- Designed and simulated a 4-DOF manipulator in **SolidWorks** and **Webots** with **ROS 2** control.
- Integrated Arduino actuation and inverse kinematic feedback for closed-loop motion.
- Applied **Design for Manufacturing (DFM)** principles and **static FEA** to ensure mechanical reliability.

Leadership & Honors

- **President & Founder, Renewable Energy Club (RENEW)** — Led sustainable energy design projects and guest seminars.
- **Facilities Chair, Mini-THON** — Managed logistics and volunteers for \$280K pediatric cancer fundraiser.
- **Honors:** CVHS Science Hall of Fame, William R. Pierce Physics Award, Lois Wolf GPA Athlete Award, Melissa Huang Scholarship.